Personal Finance Tracker– Project Report

Programming in Java Evaluated Course Project

Personal Finance Tracker

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Course: Programming in Java

University: VIT Bhopal

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1. Introduction

In daily life, people spend money on various things like food, travel, and shopping, but often do not keep a proper record of their expenses. This can make it difficult to understand where money is being spent.

This project presents a simple solution to this problem by creating a Personal Expense Tracker using Java. The application helps users record and manage their daily expenses in an organized way.

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2. Problem Statement

Many individuals, especially students, do not track their daily spending. As a result, they may lose control over their budget and face difficulty managing their finances.

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3. Objective

The main objective of this project is to develop a simple Java application that:

- Allows users to record their expenses
- Displays all recorded expenses
- Calculates the total amount spent

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4. Methodology

The project is implemented as a console-based Java program. It uses a menu-driven approach where the user can select different options. Each expense is stored as an object with two attributes:

- Category (e.g., Food, Travel)
- Amount

An ArrayList is used to store all expense records dynamically. The program runs in a loop, allowing the user to perform multiple operations until they choose to exit.

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5. Features

- Add a new expense
- View all expenses
- Calculate total expense

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6. Tools and Technologies Used

- **Programming Language**: Java
- **Concepts**: Object-Oriented Programming (OOP), ArrayList, Loops, Conditional Statements
- **IDE**: Any Java IDE (VS Code / IntelliJ / Eclipse)

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7. Challenges Faced

- Handling user input correctly
- Designing a clear menu system
- Managing data using ArrayList

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8. Learning Outcomes

Through this project, I gained a better understanding of:

- How to apply object-oriented programming in real problems
- Managing data using ArrayList
- Writing structured and user-friendly programs
- Improving logical thinking and problem-solving skills

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9. Future Improvements

- Add file handling to save data permanently
- Include date and time for each expense
- Add search and filter options
- Develop a graphical interface for better user experience

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10. Conclusion

The Personal Expense Tracker is a simple yet useful application that demonstrates how basic Java concepts can be applied to solve real-life problems. Although the project is small, it provides a strong foundation for building more advanced applications in the future.

Demo:

```
===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expense
4. Exit
Choose option: 1
Enter category: food
Enter amount: 200
Expense added successfully!

===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expense
4. Exit
Choose option: 1
Enter category: travel
Enter amount: 500
Expense added successfully!

===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expense
4. Exit
Choose option: 2

--- Expense List ---
Category: food | Amount: Rs.200.0
Category: travel | Amount: Rs.500.0

===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expense
4. Exit
Choose option: 3
Total Expense: Rs.700.0

===== Expense Tracker =====
1. Add Expense
2. View Expenses
3. Total Expense
4. Exit
Choose option: 4
Exiting... Thank you!
```